

Death Protein 5 and p53-Upregulated Modulator of Apoptosis Mediate the Endoplasmic Reticulum Stress–Mitochondrial Dialog Triggering Lipotoxic Rodent and Human β -Cell Apoptosis

Environmental factors such as diets rich in saturated fats contribute to dysfunction and death of pancreatic β -cells in diabetes. Endoplasmic reticulum (ER) stress is elicited in β -cells by saturated fatty acids. Here we show that palmitate-induced β -cell apoptosis is mediated by the intrinsic mitochondrial pathway. By microarray analysis, we identified a palmitate-triggered ER stress gene expression signature and the induction of the BH3-only proteins death protein 5 (DP5) and p53-upregulated modulator of apoptosis (PUMA). Knockdown of either protein reduced cytochrome c release, caspase-3 activation, and apoptosis in rat and human β -cells. DP5 induction depends on inositol-requiring enzyme 1 (IRE1)–dependent c-Jun NH2-terminal kinase and PKR–like ER kinase (PERK)–induced activating transcription factor (ATF3) binding to its promoter. PUMA expression is also PERK/ATF3-dependent, through tribbles 3 (TRB3)–regulated AKT inhibition and FoxO3a activation. DP5 $^{-/-}$ mice are protected from high fat diet–induced loss of glucose tolerance and have twofold greater pancreatic β -cell mass. This study elucidates the crosstalk between lipotoxic ER stress and the mitochondrial pathway of apoptosis that causes β -cell death in diabetes.